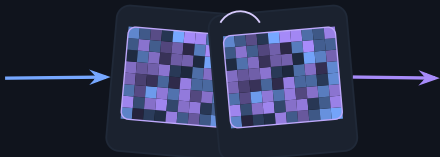


DRPE Image Encryption & Verification

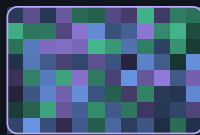
Double Random Phase Encoding for Secure Image Transformation



ORIGINAL IMAGE



DOUBLE RANDOM
PHASE MASKS



ENCRYPTED IMAGE

Mohammad Zobayer Mostafa (2305106)

■ Irfan Sadiq (2305107)

Why Encrypt Images?



ORIGINAL IMAGE

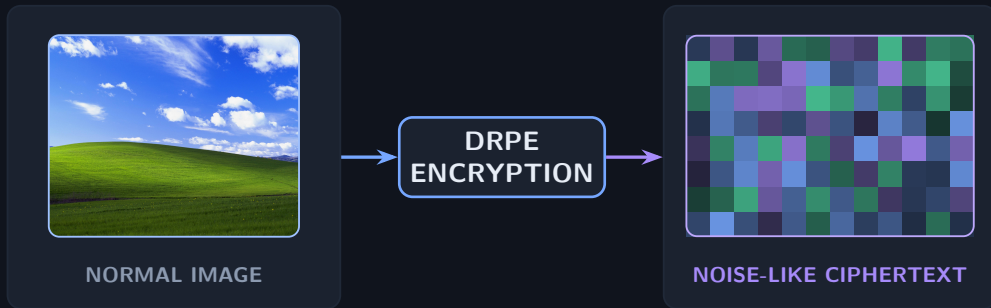
CLEARLY VISIBLE

Anyone can view it

**How can we hide an image while still allowing exact recovery
with the correct key?**

→ Our approach: **Double Random Phase Encoding (DRPE)**

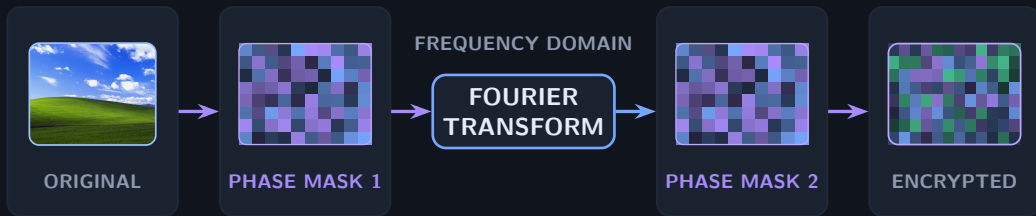
The Goal: Hide the Visual Information



The encrypted image no longer visually reveals the original content.

How DRPE Transforms the Image

SPATIAL DOMAIN



Two random phase modulations redistribute image structure into ciphertext.

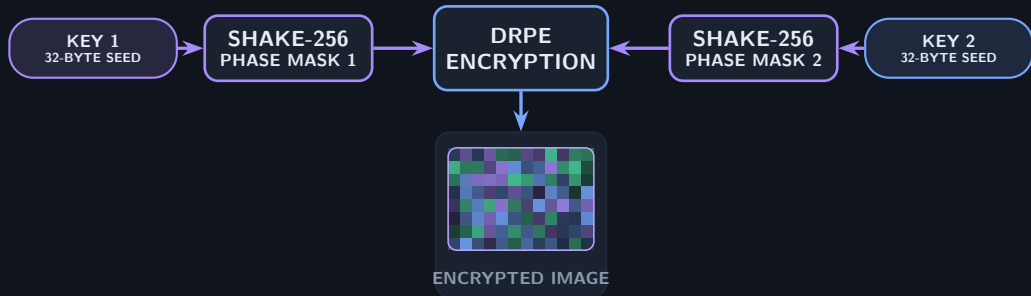
Two Independent Keys Protect Two Phase Masks



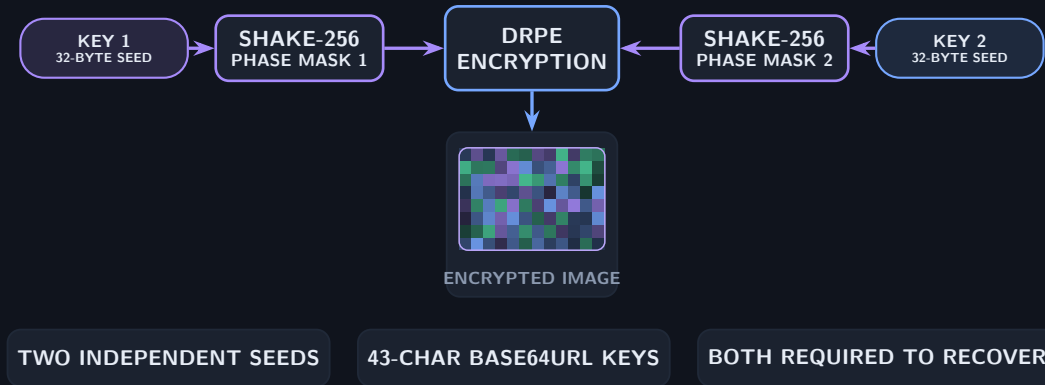
Two Independent Keys Protect Two Phase Masks



Two Independent Keys Protect Two Phase Masks



Two Independent Keys Protect Two Phase Masks



DRPE Transformation Flow

KEY IDEA Two independent random phase masks + Fourier-domain transformation $\rightarrow$ Complex-valued ciphertext.

DRPE Transformation Flow

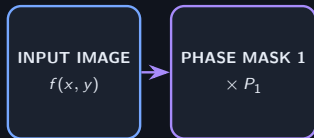
KEY IDEA Two independent random phase masks + Fourier-domain transformation $\rightarrow$ Complex-valued ciphertext.

INPUT IMAGE

$$f(x, y)$$

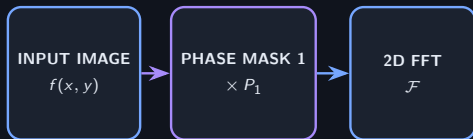
DRPE Transformation Flow

KEY IDEA Two independent random phase masks + Fourier-domain transformation $\rightarrow$ Complex-valued ciphertext.



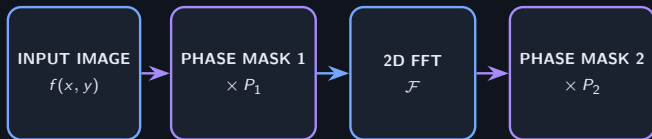
DRPE Transformation Flow

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DRPE Transformation Flow

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DRPE Transformation Flow

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DRPE Transformation Flow

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DRPE Transformation Flow

KEY IDEA Two independent random phase masks + Fourier-domain transformation $\rightarrow$ Complex-valued ciphertext.



PHASE MASK 1 — SPATIAL DOMAIN

$$P_1(x, y) = e^{i2\pi n_1(x, y)}$$

PHASE MASK 2 — FREQUENCY DOMAIN

$$P_2(u, v) = e^{i2\pi n_2(u, v)}$$

DRPE Core Equations

1 FIRST PHASE MODULATION

$$g(x, y) = f(x, y)e^{i2\pi n_1(x, y)}$$

2 FOURIER TRANSFORM

$$G(u, v) = \mathcal{F}\{g(x, y)\}$$

3 SECOND PHASE MODULATION

$$H(u, v) = G(u, v)e^{i2\pi n_2(u, v)}$$

4 ENCRYPTED COMPLEX IMAGE

$$c(x, y) = \mathcal{F}^{-1}\{H(u, v)\}$$

RECONSTRUCTION CONCEPT

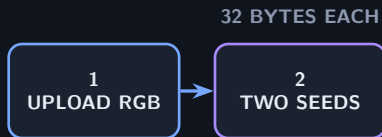
Correct $K_1, K_2 \Rightarrow$ Recover $f(x, y)$

Encryption Pipeline

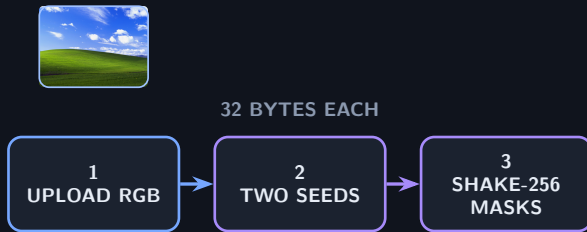


1
UPLOAD RGB

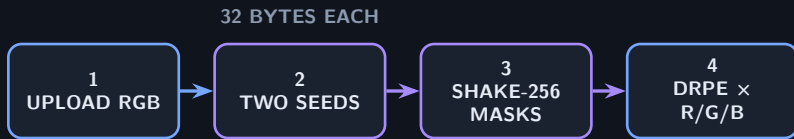
Encryption Pipeline



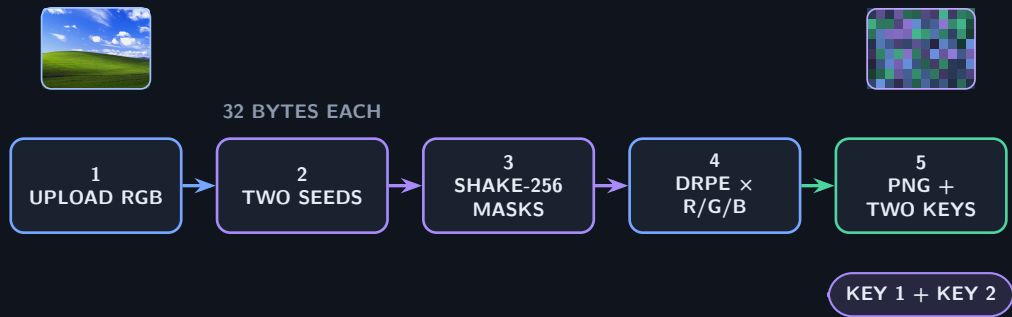
Encryption Pipeline



Encryption Pipeline

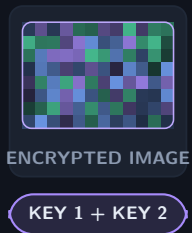


Encryption Pipeline



One clean data path: image in, encrypted PNG and two keys out.

Decryption Reverses the Transformation



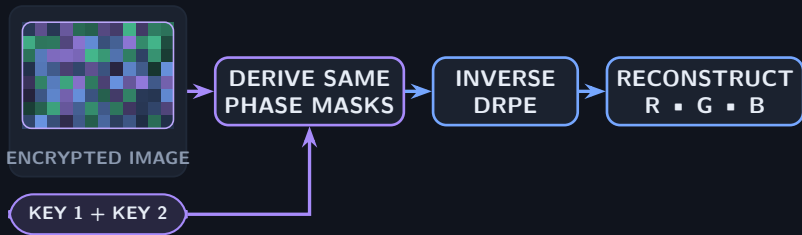
Decryption Reverses the Transformation



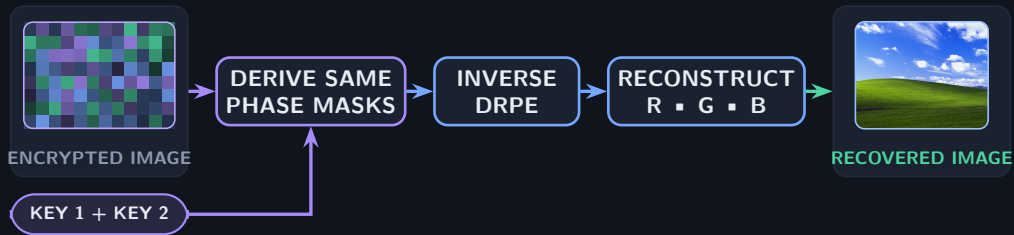
Decryption Reverses the Transformation



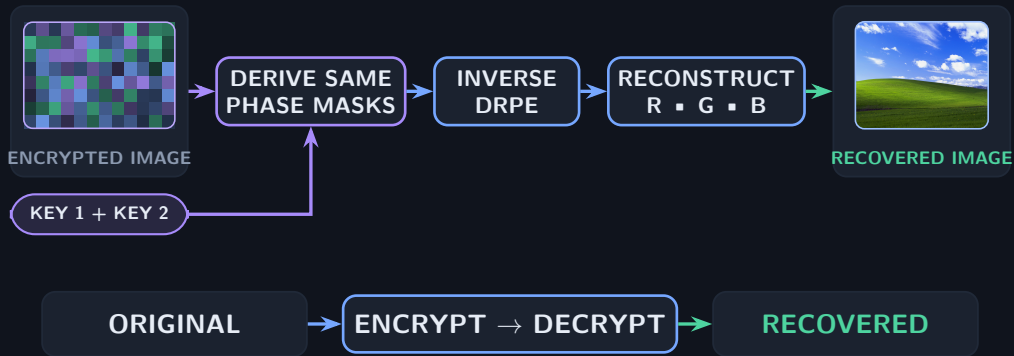
Decryption Reverses the Transformation



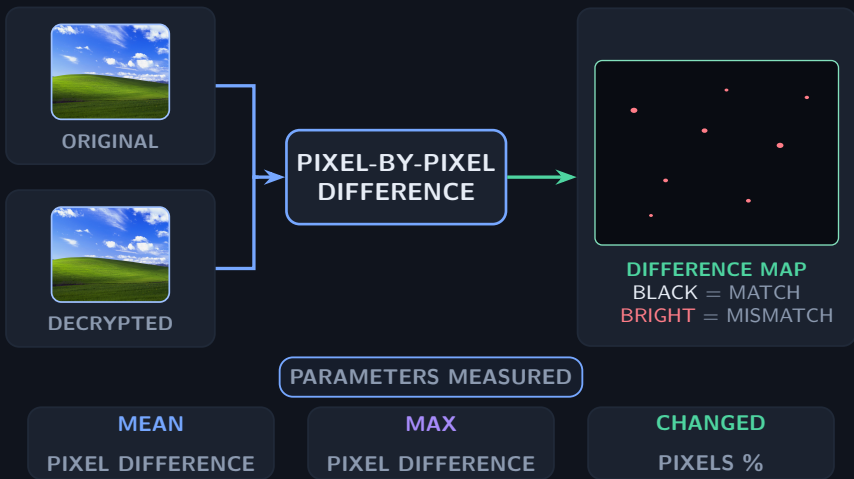
Decryption Reverses the Transformation



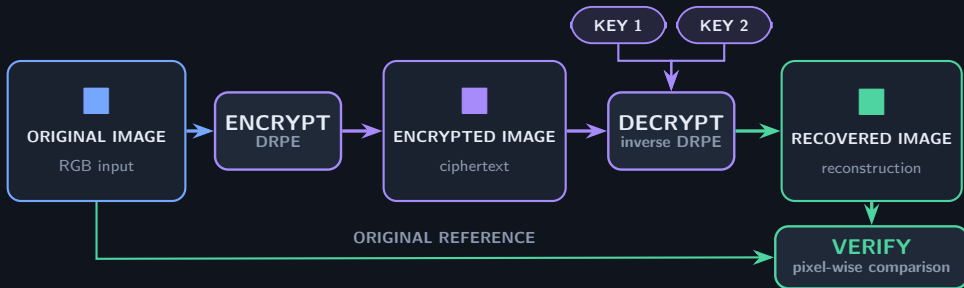
Decryption Reverses the Transformation



Did We Recover the Same Image?



Complete Encryption & Verification Workflow



Two independent keys reproduce the phase masks. Verification confirms accurate recovery.

Inside Krypt

KRYPT FRONTEND

ENCRYPT

DECRYPT

VERIFY

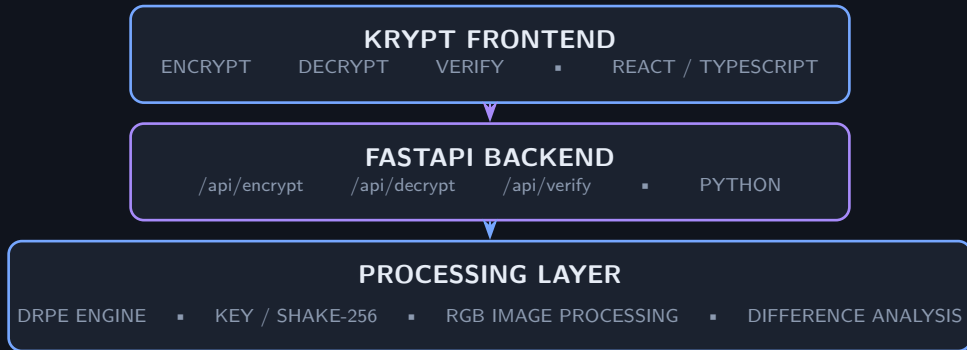


REACT / TYPESCRIPT

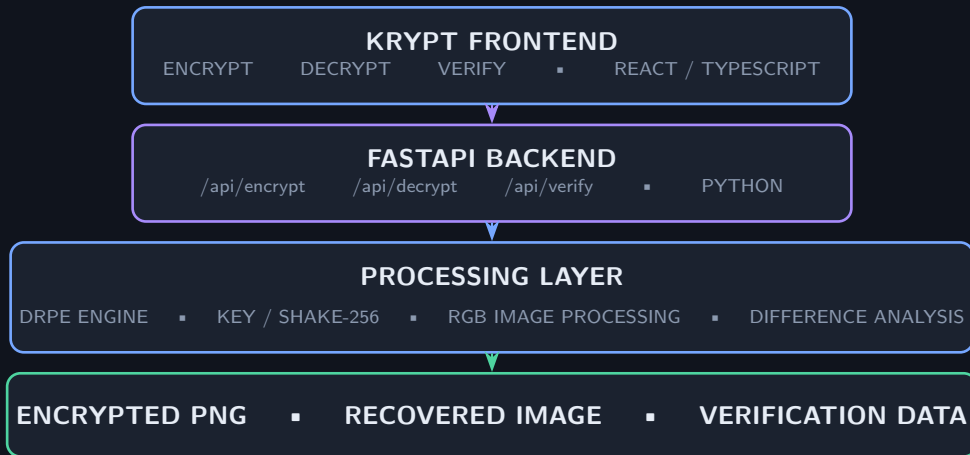
Inside Krypt



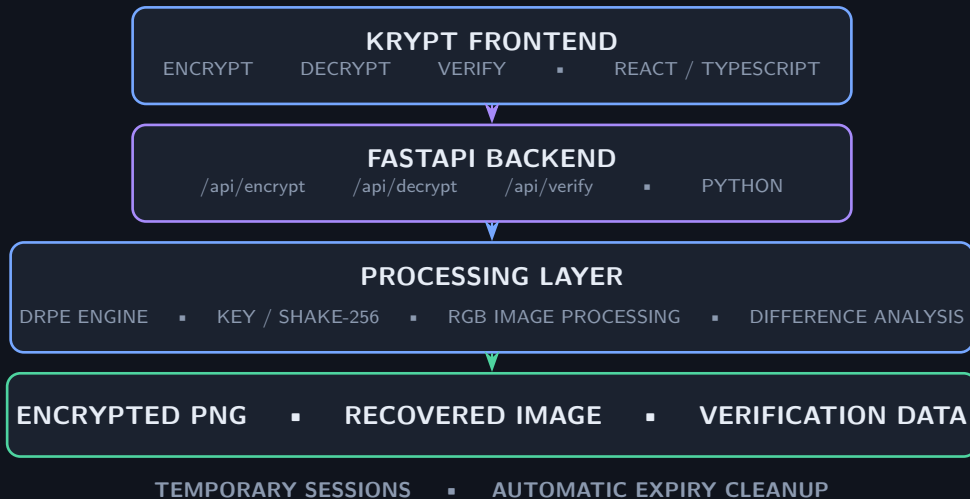
Inside Krypt



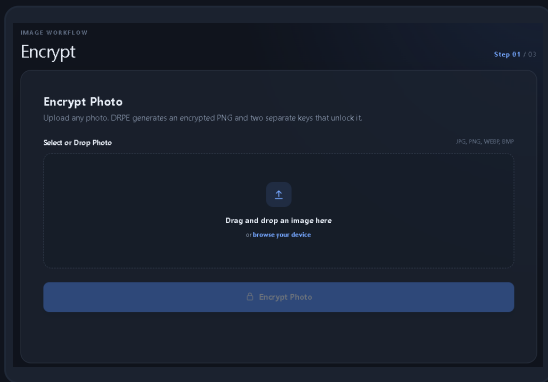
Inside Krypt



Inside Krypt



Step 1 — Upload the Image



WHAT HAPPENS

1. Choose a photo

Drag, drop, or browse for an image.

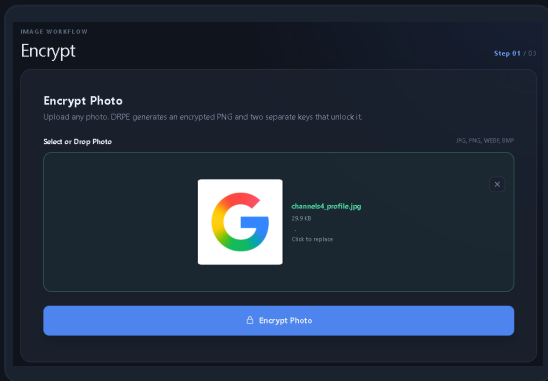
2. Preview the file

Check the selected input.

3. Start encryption

Run the Encrypt Photo action.

Step 1 — Confirm the Uploaded Photo



WHAT HAPPENS

The photo is loaded

The selected file appears in the preview.

Check the input

Confirm or replace it before processing.

Encryption is ready

The Encrypt Photo action is enabled.

Step 2 — Encrypt and Download

Encrypt Photo

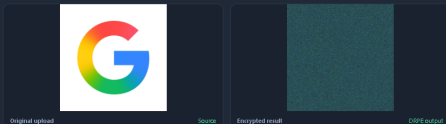
Upload any photo. DRPE generates an encrypted PNG and two separate keys that unlock it.

Encryption Complete!

Encryption successful. Save both keys; without both of them the image cannot be recovered.

Result preview

The encrypted image should look like noise. Download it before this session expires.



Download Encrypted Image

WHAT HAPPENS

RGB is encrypted

DRPE processes each color channel.

Ciphertext appears

The image becomes a noise-like PNG.

Download the result

Save the encrypted output and both keys.

Step 3 — Save Both Secret Keys

Your Two Secret Keys

Both keys are required to decrypt. Key 1 unlocks the spatial phase; Key 2 unlocks the Fourier domain phase.

● KEY 1

Copy

KEY REDACTED FOR PRESENTATION

● KEY 2

Copy

KEY REDACTED FOR PRESENTATION

Copy Both Keys

Save keys (.txt)

⚠ Save both keys now. They are not stored on the server; if either key is lost, the image cannot be recovered.

🔒 Keep both keys together with the encrypted file. Key 1 unlocks the spatial phase mask; Key 2 unlocks the Fourier-domain phase mask.

– Encrypt another photo

KEY VALUES REDACTED FOR PRESENTATION

WHAT HAPPENS

Two keys are generated

Each key is an independent secret seed.

Both are required

Together they reproduce both phase masks.

Store them safely

Keep both keys with the encrypted file.

Step 4 — Decrypt with Both Keys

Decrypt Photo

Provide the encrypted PNG and both secret keys received when encrypting.

Encrypted Image (.png)

The noise-like encrypted PNG

encrypted.png

150x118

Click to replace

Decryption Keys

Load keys from file

Key 1

.....

Key 2

.....

Decrypt Photo

KEY VALUES REDACTED FOR PRESENTATION

WHAT HAPPENS

Load the encrypted PNG

Select the ciphertext file.

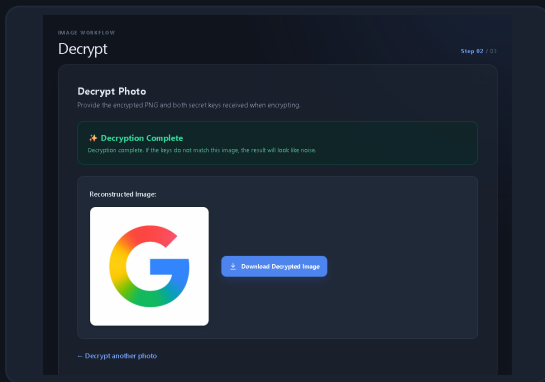
Supply Key 1 and Key 2

Both keys enable decryption.

Start reconstruction

Krypt applies inverse DRPE.

Step 5 — Recover the Image



WHAT HAPPENS

Decryption completes

RGB channels are reconstructed.

Preview the reconstruction

The recovered image is displayed.

Download or verify

Save it or continue to verification.

Step 6 — Verify the Result



WHAT HAPPENS

Compare every pixel

Original and recovered pixels are compared.

Read three measurements

Mean, maximum, and changed pixels are reported.

Receive a clear outcome

Green means match; red means mismatch.

Potential Applications



Secure Image Storage

Protect sensitive images stored on servers or local systems.



Secure Image Transmission

Transform images before sending them over an untrusted network.



Medical Image Protection

Add confidentiality for diagnostic images such as X-rays or scans.



Digital Archiving

Protect confidential image collections from unauthorized viewing.

DRPE is applicable wherever image confidentiality and controlled recovery are required.

Limitations



Key Management

Both secret keys must be stored and distributed securely.



Computational Cost

Repeated 2D FFT/IFFT operations can be costly for large images.



Ciphertext Size

Complex data needs magnitude/phase representation and metadata.



Implementation Sensitivity

Dimensions, masks, precision, and reconstruction settings must match.

Security and reliability depend on correct key handling, numerical processing, and implementation.

Thank You

Questions & Discussion

DRPE Image Encryption & Verification

Secure transformation ■ Controlled recovery ■ Measurable verification